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Willingness to Communicate and its High-Evidence Factors: A Meta-Analytic Structural Equation Modeling Approach

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Abstract

This study aimed to synthesize the structural relationships among willingness to communicate (WTC) and its high-evidence factors in second language (L2) learning contexts by adopting meta-analytic structural equation modeling (MASEM). The MASEM approach is designed to construct a structural equation model (SEM) to explain correlations between variables by pooling correlation coefficients reported in previous literature. This study integrated 44 independent samples (N = 12,094) and built a MASEM model to investigate the structural relations of WTC, its three high-evidence factors (learners' perceived L2 competence, L2 motivation, and L2 anxiety), and frequency of learners' L2 use. The results of meta-analysis successfully supported our proposed model of L2 WTC. Furthermore, among the three high-evidence factors, it was found that learners' perceived L2 competence influenced L2 WTC the most. In addition, the results of the moderator analyses indicated that learners' L2 proficiency and WTC contexts (inside vs. outside classrooms) significantly influenced the relationships between L2 WTC and its high-evidence factors. The implications of these results are discussed in further depth and detail.

Keywords: willingness to communicate, meta-analytic structural equation modeling, motivation, anxiety, competence, second language learning

This study aimed to systematically summarize the structural relationships between willingness to communicate (WTC) in second language (L2) learning contexts, and its three high-evidence (i.e., most frequently investigated) factors – namely, learners' perceived L2 competence, L2 motivation, and L2 anxiety, and the influence of these factors on learners' L2 use by employing a meta-analytic structural equation modeling (MASEM) approach.

The concept of WTC was introduced by McCroskey and Baer (1985) to examine why some people are more willing to communicate with others, and has received significant scholarly attention for its crucial role in encouraging language communication. As L2 learners' WTC is considered a gateway to frequent L2 use (MacIntyre *et al.*, 2002; Yashima *et al.*, 2004), an empirical approach has been widely used to identify significant factors that influence one's WTC. A recent meta-analysis study (Shirvan *et al.*, 2019) integrated previous findings on the subject and identified three high-evidence factors that directly and indirectly contributed to learners' L2 communication behaviors: motivation, anxiety, and perceived communication competence. Based on 60 independent samples (N = 4,794) from 22 studies, they found that the three factors were moderately correlated with L2 WTC.

Beyond the analysis of bivariate correlations, this study aimed to offer an evidence-based perspective with regard to the effects of WTC and those of WTC's factors on frequency of L2 use by providing an overall picture of causal relationships between WTC, its high-evidence factors, and frequency of L2 use. The study also aimed to help classroom educators tailor their instructions according to learners' individual characteristics. Structural equation modeling (SEM) is one of many popular statistical approaches used to test the structural relationships among constructs, and at the early stage of this research trend, MacIntyre and Charos (1996) adopted such an approach (N = 92) to investigate how different variables along with L2 WTC were structurally related to L2 use. As Cheung (2021) stated, findings in a single SEM study may not provide enough evidence, so the adoption of a MASEM approach is recommended to overcome this limitation.

MASEM is a statistical approach that combines the advantages of both meta-analysis and SEM (Cheung, 2021); thus, in the first stage, correlation matrices across previous empirical studies are *meta-analytically* combined to build a pooled correlation matrix. Therefore, although not all studies focus on correlation coefficients between all pairs of variables, this approach helps to address the missing data in the primary data. In the second stage, the pooled correlation matrix is used to fit the SEM model to test structural relations among the variables. Details of and intricacies of the MASEM approach will be further described in the methods section. We believe that MASEM is the perfect approach to achieve our research goal of testing a WTC model with its high-evidence factors.

Background

To understand why some people eagerly seek to communicate with others while others are reticent (McCroskey, 1992; McCroskey & Richmond, 1991), learners' WTC initially received attention from the L1 research community (McCroskey & Baer, 1985). Defined as a personality trait that shows similar tendencies in a variety of communication settings, the

concept of WTC has been applied to L2 communication, assuming that it operates in a similar manner to facilitate L2 communication (MacIntyre & Charos, 1996; MacIntyre *et al.*, 1998; Yashima, 2002).

WTC in Second Language Learning Contexts

In the early phase of research on WTC in L2 learning contexts (henceforth L2 WTC), McCroskey and Baer's (1985) definition of WTC, which was originally developed for WTC in L1 communication, was widely used and adapted (Mystkowska-Wiertelak & Pawlak, 2016; Peng, 2013). For example, the so-called McCroskey's WTC scale was designed to measure the amount of time a person is willing to spend communicating with three different types of receivers (i.e., stranger, acquaintance, and friend) in four communicative contexts (i.e., public, meeting, small group, and dyad), and this scale was found to show high reliability for measuring WTC, particularly in a context where learners have frequent chances to communicate in their L2 and easily encounter situations requiring L2 use (MacIntyre *et al.*, 2002).

As some L2 learners communicate in their L2s mostly in their classrooms, particularly in foreign language (FL) learning contexts, some researchers in the later phase of research on the subject (e.g., Peng, 2013) suggested that McCroskey's WTC scale needs to be revisited. Along the line, a few recent L2 WTC scales were developed by MacIntyre *et al.* (2001) and Weaver (2005), who designed their scales to better measure WTC in L2 classroom settings in addition to the previously defined four communicative contexts. For example, MacIntyre *et al.*'s (2001) WTC scale was designed to encompass learners' productive (i.e., speaking and writing) and receptive skills (i.e., comprehension and reading) in defining L2 communication, as these aspects of language learning were found to be an important part of the L2 learning process in classroom settings. Furthermore, Weaver (2005) attempted to develop MacIntyre *et al.*'s (2001) L2 WTC scale by focusing exclusively on learners' speaking and writing skills and revising questionnaire items to better capture L2 WTC in L2 classroom settings.

Such approaches could be realized based on a belief that L2 WTC is not merely a "simple manifestation of WTC in the L1" (MacIntyre *et al.*, 1998, p. 546) in that L2 WTC interacts with L2 related variables, including social and psychological factors, which may not be correlated with L1 counterparts. To give an extreme example, one early study even found a negative correlation between L1 WTC and L2 WTC (Charos, 1994). Hence, MacIntyre and Charos (1996) integrated Gardner's (1985) socio-educational model to demonstrate that L2 WTC was influenced by L2 motivation, L2 anxiety, and perceived L2 competence, which corresponded with previous findings in the L1 context associating L1 WTC with L1 self-esteem, L1 perceived communication competence, and L1 communication apprehension.

Likewise, based on the belief that L1 WTC and L2 WTC appeared to operate independently with separate sets of factors in L1 and L2 contexts, respectively, MacIntyre *et al.*'s (2001) and Weaver's (2005) WTC scales were developed, and most quantitative studies of L2 WTC have utilized either of the two scales or adapted them to suit their own research contexts (e.g., Peng & Woodrow, 2010).

WTC and its High-Evidence Factors

MacIntyre and Charos' (1996) conceptual model of L2 WTC suggested the existence of complex relationships between WTC, factors influencing WTC, and L2 use. Subsequent researchers have investigated this model mostly by using quantitative methods to acquire empirical evidence, and attempted to improve the model by identifying the relationships between L2 WTC and especially social psychological variables, such as personality, motivation (e.g., Hashimoto, 2002; Khajavy *et al.*, 2016), international posture (e.g., Yashima *et al.*, 2004), self-confidence (e.g., Fallah, 2014; Yashima, 2002), attitudes toward L2 and perceived communicative competence (e.g., Khajavy *et al.*, 2016), anxiety in speaking the L2 (e.g., Denies *et al.*, 2015; MacIntyre & Charos, 1996), and L2 grit (Lee, 2020; Lee & Lee, 2020).

These studies on the relationships between L2 WTC and social psychological factors have investigated how the interplay between these constructs operates in different social, cultural, and educational contexts. For example, with respect to L2 motivation, it has been well established that learners with higher L2 motivation are likely to have higher L2 WTC in classrooms (Cha & Kim, 2013; Dörnyei & Kormos, 2000; Hashimoto, 2002; Khajavy *et al.*, 2016; Lee & Lee, 2020). Moreover, perceived communicative competence has been found to positively influence L2 WTC not only in classrooms (Hashimoto, 2002; Khajavy *et al.*, 2016; Yashima, 2002), but also outside classrooms (Lee & Lee, 2020; Pyun *et al.*, 2014; Samimy & Tabuse, 1992). When it comes to learners' anxiety, a negative correlation between L2 speaking anxiety has been corroborated by a plethora of studies in both SL and FL contexts (Cha & Kim, 2013; Denies *et al.*, 2015; Hashimoto, 2002; Khajavy *et al.*, 2016; MacIntyre & Charos, 1996). More specifically, several studies including those conducted by Horwitz (2016) and Park (2014) have demonstrated that the degree of learners' L2 anxiety may vary across different cultures and learning contexts. In the same vein, Lee (2020) identified a stronger correlation between L2 anxiety and L2 WTC in classrooms than outside classrooms and digital environments. Other relatively less studied factors, such as international posture, attitudes toward L2, and L2 grit, have been found to positively influence L2 WTC in FL classrooms (Khajavy *et al.*, 2016; Lee & Lee, 2020; Yashima *et al.*, 2004).

WTC and its Contribution to L2 Use

MacIntyre *et al.* (1998) proposed a heuristic blueprint for the L2 WTC construct and provided a conceptual framework of its influence on L2, with WTC being associated with diverse linguistic, contextual, communicative, and social-psychological factors (Figure 1). This six-layer pyramid model suggests that L2 WTC is influenced not only by "personality-based, trait-like predisposition" (McCroskey & Richmond, 1991, p. 134) but also diverse contextual-state variables and can be conceptualized as "a readiness to enter into discourse at a particular time with a specific person or persons, using a L2" (MacIntyre *et al.*, 1998, p. 547). Based on this model, L2 WTC has received significant scholarly attention as an important construct affecting L2 communication behavior (Yashima *et al.*, 2018), including individual trait-like, contextual state-like, and dynamic perspectives (Lee & Lee, 2020).

WTC was considered a crucial factor in L2 learning, as MacIntyre *et al.* (1998) described it as an immediate predecessor to L2 use. Along with the paradigm shift to communicative language teaching (CLT) in L2 learning (Khajavy *et al.*, 2016), WTC received attention, as it directly affects L2 use and achievement and is one of the strongest predictors of L2 communication frequency. Clément *et al.* (2003) investigated the relationship between individual and contextual variables of university students in Canada, confirmed that WTC significantly predicted learners' L2 use frequency, and examined a triangular relationship among confidence (i.e., competence), WTC, and L2 use based on the integration of Clément's (1980) social context model of L2 competence and MacIntyre *et al.*'s (1998) WTC model. MacIntyre and Charos (1996) tested a hybrid model combining Gardner's (1985) socio-educational model and MacIntyre's (1994) WTC model to examine the relationship among variables influencing L2 communication and discovered that L2 communication frequency was affected by WTC, motivation, and perceived competence. Yashima *et al.* (2004) examined the variables affecting WTC and L2 use among Japanese high school students and found that WTC led to frequent L2 communication inside and outside classrooms.

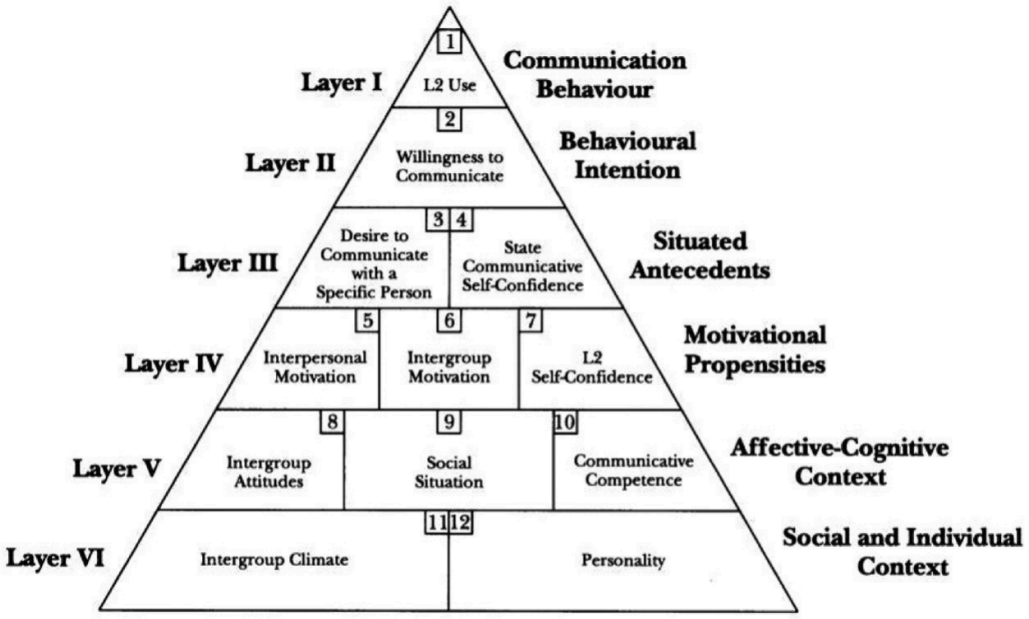


Figure 1. Macintyre *et al.*'s (1998; reproduced with permission) heuristic model of variables influencing L2 WTC.

Previous Meta-Analyses on WTC in L2 Contexts

Shirvan *et al.* (2019) conducted a meta-analysis to integrate previous findings on L2 WTC and its high-evidence factors and identified learner motivation, anxiety, and perceived communication competence as high-evidence variables. Learner motivation is one of the most researched learner factors (Al-Hoorie & MacIntyre, 2019; Dörnyei, 1998; Gardner, 2010; Gardner & MacIntyre, 1993) and considered important in influencing L2 WTC (Peng & Woodrow, 2010). It is widely accepted that L2 learners with high motivation are more willing to communicate with others in L2 (Knell & Chi, 2012; Liu & Park, 2012; Peng & Woodrow, 2010). Shirvan *et al.* (2019) identified nine correlation coefficients between L2 motivation and WTC, with an overall mean correlation of .37.

In addition, anxiety has been found to be crucial in L2 learning (Horwitz, 1986; Horwitz *et al.*, 1986; MacIntyre & Gardner, 1989; MacIntyre & Gardner, 1991). Many studies demonstrated a negative impact of anxiety on L2 WTC (Gardner & MacIntyre, 1993; Hashimoto, 2002; Yashima, 2002), indicating that language learners who are anxious about using L2 tend to be reticent in L2 communication. The meta-analytic results revealed 12 correlation coefficients between L2 anxiety and L2 WTC, with an overall mean correlation of − .29.

McCroskey and Richmond (1991) conceptualized perceived communication competence as learners' perception and evaluation of their competence, which has been found to influence L2 learning and communication (Clément *et al.*, 1994). A number of studies asserted that perceived communication competence is one of the strongest predictors of L2 WTC, indicating that L2 learners with higher perceived communication competence are more willing

to communicate in L2 (e.g., MacIntyre & Charos, 1996; MacIntyre *et al.*, 1998). Shirvan *et al.* (2019) found 11 correlation coefficients between perceived L2 competence and L2 WTC, with overall mean correlation of .48.

In sum, Shirvan *et al.* (2019) discovered three high-evidence variables that were moderately correlated with L2 WTC and identified perceived communication competence as the strongest variable in predicting L2 WTC. As, to the best of the authors' knowledge, Shirvan *et al.* (2019) was the first such meta-analytic effort, further research is required to understand the *structural* relationship, beyond bivariate correlations, between these variables, which the present study aimed to address.

Present Study

In addition to the importance of the three high-evidence factors, researchers have explored their structural relationships with L2 WTC (e.g., Hashimoto, 2002; MacIntyre & Charos, 1996; Peng & Woodrow, 2010). Hashimoto (2002) revised MacIntyre and Charos' (1996) path model and discovered that perceived competence and anxiety directly influenced WTC while WTC affected motivation and L2 communication frequency in the Japanese ESL context. Peng and Woodrow (2010) revealed the direct influence of communication confidence and indirect effect of motivation on L2 WTC.

As Shirvan *et al.*'s (2019) meta-analysis summarized the relationships between L2 WTC and its high-evidence factors and provided a reliable and accumulated perspective, extant empirical findings on the structural relationships between WTC and its high-evidence factors should be integrated. As such, this study adopted a MASEM approach, which was designed to create an SEM model based on a pooled correlation matrix by integrating raw correlation coefficients collected from the literature (Jak & Cheung, 2020).

The present study is expected to extend Shirvan *et al.*'s (2019) meta-analytic effort and provide a deeper understanding of the *structural* relationships between L2 WTC and the three high-evidence factors. In addition, the MASEM model examined the influence of WTC and its three factors on L2 use (Figure 2). Furthermore, the moderating effects of several learner factors on the structural relationships between the target variables were analyzed.

It should be noted here that as we briefly mentioned above in addition to quantitative studies, including the present study, investigating the relationships between L2 WTC and other factors, such as individual, cognitive, and social psychological variables, there have also been a significant number of qualitative studies (e.g., Cao, 2011; Cao, 2014; Cao & Philp, 2006; Kang, 2005; Peng, 2012) that have delved into understanding how L2 WTC changes depending on a variety of individual, situational, and contextual factors. For example, by

implementing qualitative methods, including interviews, Kang (2005) conceptualized that L2 WTC emerges and fluctuates in the process of dynamic interaction with learners' psychological conditions (i.e., excitement, responsibility, and security) and situational variables (i.e., interlocutor, topic, and conversational context). In a similar vein, Cao and Philp (2006) found that L2 WTC is significantly affected by contextual factors, such as familiarity with topics and interlocutors. Adopting an ecological perspective, Cao (2011) and Peng (2012) identified the joint effect of various individual (e.g., self-confidence, personality), classroom environmental (e.g., topics, interlocutors), and linguistic (e.g., L2 proficiency) variables on L2 learners' WTC in classrooms.

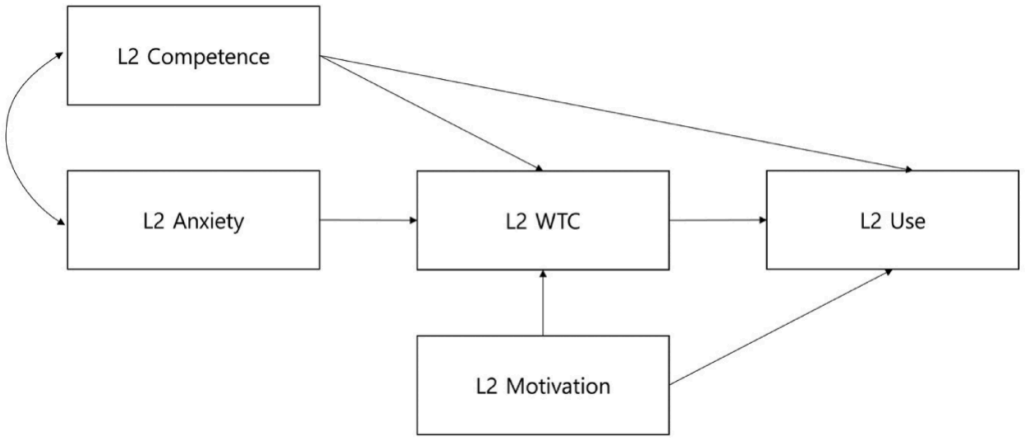


Figure 2. Proposed SEM model adapted from MacIntyre and Charos' (1996) L2 WTC model.

Furthermore, recent studies have started to explore the dynamic nature of L2 WTC as dynamic approaches to second language acquisition (SLA) are of growing interest in the field of applied linguistics. The L2 WTC literature has in fact taken a new turn with the introduction of the complex dynamic systems theory (CDST) by Larson-Freeman and Cameron (2008). Influenced by the conceptualization of language learning as a dynamic system in which a variety of individual properties and situational variables interact with each other, and ultimately influence language learning (Larson-Freeman & Cameron, 2008), many studies have started to explore the rapid and moment-by-moment changes of L2 WTC to fully acquire a comprehensive blueprint of WTC in language learning (Ducker, 2021; MacIntyre, 2012; MacIntyre & Doucette, 2010; MacIntyre & Gregersen, 2021; Mystkowska-Wiertelack & Pawlak, 2016; Nematizadeh, 2021).

The recent trend of focusing on L2 WTC as an emergent and dynamic variable does not necessarily mean that previous studies on L2 WTC as a stable and enduring variable are outdated or no longer valid. Rather, the two different approaches to L2 WTC are complementary and have the potential to provide a comprehensive understanding of the subject. That is, L2 WTC can be understood to have both a trait (individually different) and

state (context-sensitive) in L2 communication (Zhang *et al.*, 2018); the former indicates one's general communicative propensity, while the latter changes and fluctuates on a momentary basis in a variety of communicative contexts. The focal theme of the present study is the trait-like qualities of L2 WTC as a relatively stable individual characteristic as this study aims to examine the relationships between L2 WTC and its high-evidence factors over a longer period (MacIntyre, 2020).

Methods

Literature Search

Previous empirical studies were located using a systematic procedure and the study list was refined to exclude studies that did not meet the inclusion criteria. The inclusion criteria were: (a) studies should be written in English, (b) studies should be conducted in L2 learning contexts, (c) studies should report raw correlation coefficients (r) between any of the five target variables, (d) studies should report separate data when L1 and L2 learners are included, and (e) studies should report sample sizes. Figure 3 illustrates the steps conducted following the preferred reporting items for systematic reviews and meta-analyses (PRISMA; Page *et al.*, 2021).

Empirical studies were systematically identified using keyword (i.e., database and manual searches) and forward/backward searches up to the year 2021. For the former, several databases, Scopus, ProQuest, Web of Science, and relevant journals, The Modern Language Journal, Applied Linguistics, Language Learning, Studies in Second Language Acquisition, TESOL Quarterly, were used with the following combination of keywords for the abstract field: "willingness to communicate" AND ["second language" OR "foreign language"] AND ["motivation" OR "anxiety" OR "competence"]. For the latter, reference lists of the identified studies were reviewed for the backward search and Google Scholar's "Cited by" function was used for the forward search. After eliminating duplicates, 97 studies were identified.

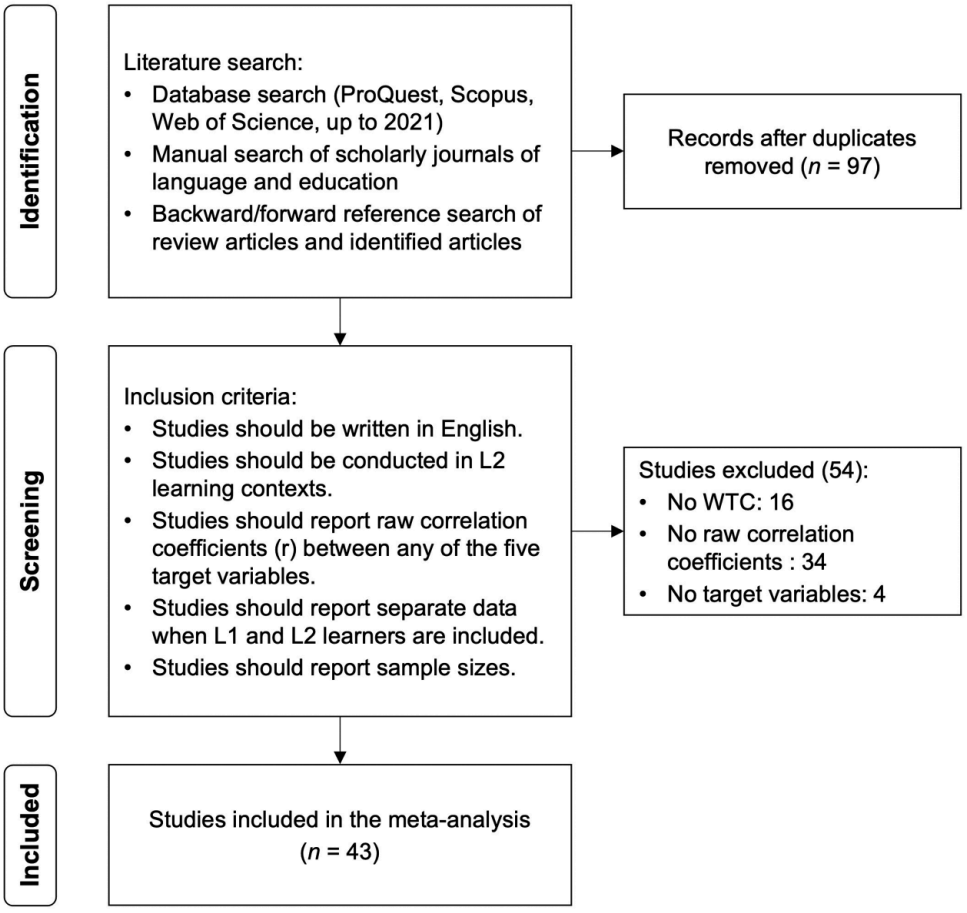


Figure 3. PRISMA flow diagram (adapted from Page *et al.*, 2021).

A thorough review was conducted to check if the studies met the inclusion criteria. A total of 43 studies were included after excluding 54 studies (16 studies were irrelevant to WTC, 34 studies were qualitative studies on WTC and did not include correlation coefficients, and four studies did not include target variables of interest). MacIntyre *et al.* (2003) included two independent samples based on treatment type. Therefore, a total of 44 samples (N = 12,094) from 43 studies were included in the present analysis.

Coding Procedure

After finalizing the dataset, the samples were coded by checking sample sizes, summarizing correlation matrices, and identifying moderator variables.

Correlation Matrices. After a review of each study, reported raw correlation coefficients among any of the five target variables – L2 WTC, L2 motivation, L2 anxiety, perceived L2 competence, and L2 use – were transcribed. Although the collected studies commonly focused on learners' WTC in English learning contexts, there was some variation in terms of definitions and examples of measurements for the four independent variables. We found that the majority of these studies (e.g., Fallah, 2014; Ghonsooly *et al.*, 2012; Hashimoto, 2002;

Yashima, 2002) used McCroskey's (1992) WTC scale as the most common instrument to measures learners' WTC. This scale deals with participants' propensity to initiate communication in L2 with three types of receivers (strangers, acquaintances, and friends) in four communication contexts (public speaking, talking in meetings, talking in small groups, and talking in dyads), where a typical L2 (not FL) learning situation (i.e., outside of a classroom) is considered. As previously noted, to better capture more diverse L2 learning settings, particularly classroom contexts, some recent studies (e.g., Choi, 2021; Khajavy *et al.*, 2016; Peng & Woodrow, 2010; Zarrinabadi *et al.*, 2021) used WTC scales developed by MacIntyre *et al.* (2001) and Weaver (2005) (i.e., inside classrooms). For example, Peng and Woodrow (2010) adapted these scales to suit their own research contexts and objectives to measure learners' tendencies to start L2 communication with their teachers, peers, and group mates on topics such as self-introduction, short speeches about their hometowns, pronunciation, and the meaning of words in L2.

Regarding motivation, we found a variety of different measurements across the primary studies, as this is one of the most explored constructs in the literature (Al-Hoorie & MacIntyre, 2019; Dörnyei, 1998; Gardner, 2010; Gardner & MacIntyre, 1993). Early studies utilized the concept of integrative motivation (i.e., attitudes toward the learning situation), largely based on Gardner's (1985) socio-educational model. The focus of motivation research shifted in the 1990s, and many studies during this period were influenced by Deci and Ryan's (1985) self-determination theory (SDT). This construct was seen as either intrinsic motivation, defined as one's desire to achieve language learning, or extrinsic motivation, which emerges from external factors, such as receiving praise or escaping penalty (Noels *et al.*, 2000). Following Dörnyei's (2009) L2 motivational self-system, studies proposed the ideal L2 self (L2-specific attributes of learners' ideal self when they can master L2) and ought-to L2 self (L2-specific attributes that learners believe they ought to achieve) to measure learners' motivation from cognitive and psychological perspectives (Liu & Park, 2012).

With regard to anxiety, we found that studies used a variety of measurements with a common goal of assessing learners' apprehension. For example, Horwitz *et al.* (1986) developed the foreign language classroom anxiety scale (FLCAS) to evaluate anxiety, including three components: communication apprehension, test anxiety, and fear of negative evaluation. In addition, McCroskey *et al.* (1987) created a communication apprehension scale to measure communication anxiety with friends, acquaintances, and strangers in situations such as dyads, groups, meetings, and in public. This scale was also adapted by MacIntyre and Charos (1996) and Yashima (2002) to measure learners' anxiety in L2 classrooms and L2 use. Lastly, concerning L2 learners' perceived communicative competence, we found that a scale proposed by McCroskey *et al.* (1987) and its variations were widely used in the primary studies.

MacIntyre and Charos (1996) and MacIntyre *et al.* (2002) utilized items adapted from McCroskey *et al.* (1987) to measure the participants' perceived competence in L2. Yashima (2002) used items from MacIntyre and Charos (1996) to measure participants' self-evaluated competence in L2 communication using the same receivers and communication contexts used by McCroskey (1992). MacIntyre *et al.* (2002) utilized 12 items from MacIntyre and Charos (1996), which were adapted by McCroskey *et al.* (1987).

A total of five independent samples (out of 44 samples, or 11%) were found to have multiple correlation coefficients between constructs of interest. To avoid issues of dependency, the correlation coefficients were averaged, and this approach seemed most appropriate at least in the case of three (Khodadady & Khajavy, 2013; Liu, 2018; Peng, 2015) of the five samples where multiple sub-constructs (or factors) were employed for precision. However, in two cases (MacIntyre & Blackie, 2012; Wu & Lin, 2014) where multiple (somewhat) distinguishable measurements were obtained, the adoption of this approach could be problematic. For instance, as shown in Table 1, Wu and Lin (2014) measured four variables – instrumental motivation, integrative motivation, speaking anxiety, and WTC – and the correlation coefficient matrix contained six correlations between these variables (see Table 1, top). Although the two measures for learners' overall motivation (i.e., instrumental and integrative motivation constructs) were different measurements, for statistical dependency, they were averaged to compute a single construct for the correlation matrix (Table 1, bottom). In this case, the average correlation coefficient between motivation and anxiety was − .65 based on two correlation coefficients ranging from − .61 to − .69, and the average correlation coefficient between motivation and WTC was .63 based on two correlation coefficients ranging from .60 to .65. Supplementary Material 1 includes details about the numbers of pairs we had to average and differences between the correlation coefficients across the primary studies, and it is evident that the averaged pairs had the same directions and showed little differences in size between the targeted raw coefficients.

Table 1. Correlation Matrix for Multiple Measures for Variables of Interest (e.g., Wu & Lin, 2014).

[Original correlation matrix]				
	1.	2.	3.	4.
1. Instrumental motivation	–			
2. Integrative motivation	.68	–		
3. Speaking anxiety	−.69	−.61	–	
4. Willingness to communicate	.60	.65	−.67	–
↓				
[Revised correlation matrix for the MASEM approach]				
	1 + 2.	3.	4.	
1 + 2. Motivation	–			

3. Speaking anxiety	−.65	–	
4. Willingness to communicate	.63	−.67	–

We acknowledge that this approach may cause potential interpretive difficulties as different measurement types have different emphases and different psychometric properties. Nevertheless, we decided to follow the same path as previous MASEM studies (e.g., Peng *et al.*, 2021; Quinn & Wagner, 2018) as it is not possible to include and review every different theory in detail due to practical limitations. Consequently, we were able to code our correlation matrices for a total of five variables based on the collected raw correlation coefficients from 44 independent samples (N = 12,094). Table 2 displays the number of raw correlation coefficients for pairs of constructs of under examination. The numbers ranged from five (11%) to 31 (70%), indicating that the extent of missing data was between 30% and 89%.

Moderator Variables. After coding raw correlation coefficients and sample sizes, variations were identified in several features, such as (a) L2 learning contexts, (b) L1-L2 similarities, (c) WTC contexts, and (d) learners' age and (e) L2 proficiency levels. Supplementary Material 1 summarizes the differences between the samples regarding these moderator variables. First, with regard to contextual features, approximately 70% (31 out of 44) were conducted in foreign language (FL) learning contexts, while approximately 30% (13 out of 44) were conducted in second language (SL) learning contexts. In terms of L1-L2 similarities, it was found that students in five samples had L2s that were linguistically related to their L1s. According to the results of Beaufils and Tomin's (2020) genetic proximity calculator between two languages, this variable was coded as no relation (78–100) or related (1–77). Different WTC contexts were coded. By using different WTC scales across the primary studies, we identified that there were two contexts of WTC: inside (MacIntyre *et al.*'s [2001] and Weaver's [2005] WTC scales) and outside classrooms (McCroskey's WTC scale). We found that 22 studies (50%) focused on WTC outside the classroom, 19 studies (43%) looked at WTC inside the classroom, and three studies (7%) did not include WTC in their analyses. Second, related to learner features, age was coded as children or adolescents/adults. Only two samples (5%) were conducted with children, while the remaining 42 samples (95%) included adolescents or adults. Learners' L2 proficiency was coded as basic or beyond basic. Six samples (14%) did not provide relevant information while only three samples included low proficiency learners (7%). These five moderator variables were coded and included in further data analyses, as they could influence the structural relationships between the target variables in the SEM model.

Table 2. Number of Correlation Coefficients from Previous Studies.

	1.	**2.**	**3.**	**4.**	**5.**
1. L2 WTC	44 (12,094)				

	1.	2.	3.	4.	5.
2. L2 Motivation	28 (9,214)	44 (12,094)			
3. L2 Anxiety	31 (9,479)	24 (7,745)	44 (12,094)		
4. L2 Competence	25 (7,800)	17 (5,529)	22 (6,594)	44 (12,094)	
5. L2 Use	8 (1,798)	6 (1,529)	5 (1,271)	5 (1,271)	44 (12,094)

Note. Values in parentheses indicate total sample sizes for correlation coefficients collected from primary studies.

Data Analysis

This study used the *metaSEM* package (Cheung, 2015; version 1.2.5.1) in *R* (version 4.0.2). First, a pooled correlation matrix was created by integrating the collected raw correlation coefficients from 44 independent samples. As most samples did not have correlations between all five variables of interest (Table 2), the MASEM approach, employs the full information maximum likelihood (FML) estimation method to estimate a likelihood function with the available data on the assumption that the studies missed the values without a specific reason, that is, at random. Second, an SEM model was created using the matrix to check whether the collected data fit the model using the chi-square test (χ^2 test; $p > .05$ for acceptable fit) and four widely used model fit indices (Lee *et al.*, 2020): (a) root mean square error of approximation (RMSEA; < .08 for acceptable fit), (b) root mean square residual (SRMR; < .08 for acceptable fit), (c) comparative fit index (CFI; > .90 for acceptable fit), and (d) Tucker–Lewis index (TLI; > .90 for acceptable fit). Third, when the baseline model (i.e., without a moderator) was established and passed the model fit indices, six analyses were conducted for each of the six moderators to investigate how the potential moderator variables influenced path coefficients between the variables of interest. When a moderation effect was detected among the six path coefficients, follow-up analyses were conducted with only significant paths included to investigate the moderator effects for a higher precision (Hox, 2010). The statistical significance of moderating effects was measured by any significant difference between a model with the moderator and the baseline model from the results of a likelihood ratio test ($p < .05$ for significant moderating effects).

Results

Overall Analysis

Table 3 presents the pooled correlation matrix computed from the raw correlation coefficients of the 44 samples. The average correlation coefficients in this matrix were calculated considering different sample sizes of each sample by assigning corresponding weights (Jak &

Cheung, 2020). The results indicated that learners' motivation and competence were positively correlated with WTC and anxiety was negatively correlated with WTC, which was correlated with L2 use frequency.

Based on this pooled correlation matrix, the MASEM approach demonstrated whether the data fit the proposed SEM model. Figure 4 shows the results. The baseline MASEM model without any moderator successfully passed the chi-square test (χ^2 (1) = 3.357, p = .067) and the four model fit indices: CFI = .993, TLI = .932, RMSEA = .014, and SRMR = .031.

The results shown in Figure 4 indicate that, first, the three high-evidence factors – learners' perceived communication competence (β = .357, SE = 0.040, z = 8.983, p = .000), motivation (β = .187, SE = 0.025, z = 7.500, p = .000), and anxiety (β = –.114, SE = 0.036, z = −3.187, p = .001) – had statistically significant direct contributions to L2 WTC. To compare the magnitudes of each path coefficient, we further conducted a likelihood ratio (LR) test for statistical comparisons. The results indicated that there was a significant difference across the path coefficients for these three factors (χ^2 (2) = 89.612, p = .000), and that competence (competence > motivation; χ^2 (1) = 8.601, p = 0.003) had the largest contribution, followed by motivation (motivation > anxiety; χ^2 (1) = 38.622, p = .000) and anxiety (competence > anxiety; χ^2 (1) = 63.125, p = .000).

When it comes to L2 use as a dependent variable, we found that competence (β = .362, SE = 0.144, z = 2.505, p = .001), motivation (β = .179, SE = 0.070, z = 2.546, p = .001), and L2 WTC (β = .178, SE = 0.090, z = 1.980, p = .048) had statistically significant direct contributions to L2 use. To statistically compare the magnitudes of each path coefficient, we conducted an LR test once again, but the results indicated that there were no significant differences across these three path coefficients (χ^2 (2) = 1.166, p = .558. We speculate that this was due to the limited number of correlation coefficients for L2 use collected from previous studies (see Table 2).

The direct associations between the factors are clearly indicated by arrows and their coefficients in Figure 4. However, the figure does not fully represent indirect effects among the factors; therefore, we tabulated estimates for direct, indirect, and total effects of the constructs in the baseline MASEM model (see Table 4). The results indicated that all three high-evidence factors had indirect effects on L2 use mediated by L2 WTC. Although these indirect effects were not statistically significant, in case of learners' competence and motivation (except in the case of anxiety) they contributed to their total effects on L2 use, allowing us to obtain more precise estimates on how these variables affected the frequency of learners' L2 use. On the contrary, we were able to identify that learners' anxiety did not affect

their L2 use either directly (no direct connection) or indirectly (β = –.020, SE = 0.069, *z* = –.296, *p* = .767), although learners' L2 WTC already affected by learners' anxiety would definitely influence L2 use.

Table 3. Pooled Correlation Matrix.

	1.	2.	3.	4.	5.
1. L2 WTC	–				
2. L2 Motivation	.336	–			
3. L2 Anxiety	−.311	−.281	–		
4. L2 Competence	.467	.327	−.400	–	
5. L2 Use	.392	.341	–.307	.429	–

Note. All correlation coefficients included in the table were statistically significant ($p < .05$).

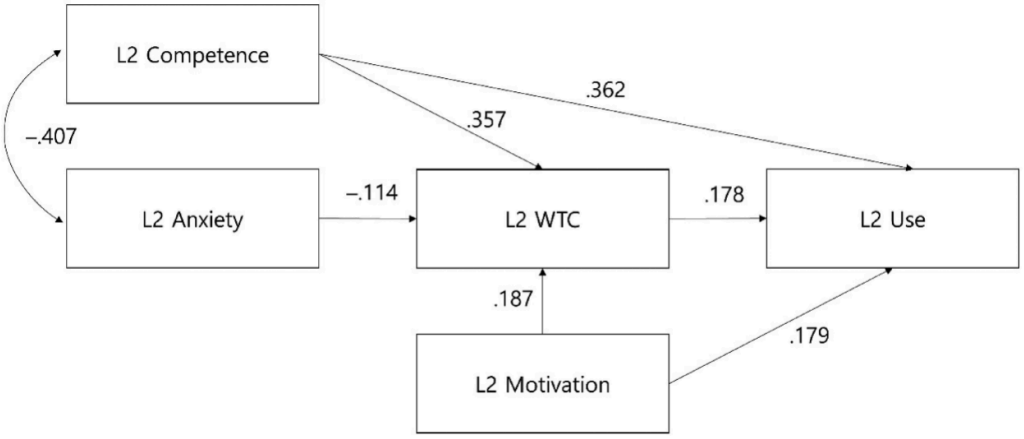


Figure 4. Baseline MASEM model without any moderators. Note. The solid lines in the figure represent the statistically significant path coefficients. The values of the path coefficients are the correlation coefficients between the variables.

Moderator Analysis

A total of five moderator variables – language learning contexts (SL vs. FL), L1-L2 relation (related vs. not related), WTC contexts (inside vs. outside classrooms), learners' age (children vs. adolescents/adults), and learners' L2 proficiency (basic vs. beyond basic) – were coded, and a series of analyses for each moderator were conducted to investigate how these variables influenced the complex relationships among L2 WTC and its high-evidence factors. Table 5 represents the results of the moderator analyses.

As the first step, each moderator analysis was conducted by statistically comparing (by means of an LR test) the baseline MASEM model with a nested MASEM model with a moderator included on each of the path coefficients. The results indicated that three moderator variables – language learning contexts (SL vs. FL), L1-L2 relation (related vs. not related), and learners' age (children vs. adolescents/adults) – did not show statistically significant improvements

when added to the baseline model (*p*s > .05). We found that the "WTC context" (χ^2 (6) = 11.163, p = .025) and "learners' L2 proficiency" (χ^2 (6) = 14.963, p = .021) moderators had statistical significance in this step.

For the second step, considering that the significance of these two moderators (i.e., the WTC context and learners' L2 proficiency) did not mean that every path coefficient was significantly influenced by these moderators, we conducted follow-up analyses with moderators included on the selected path coefficients, which had statistical significance in the earlier step. For the "WTC context" moderator, two paths – one between L2 anxiety and L2 WTC, and another between perceived L2 competence and L2 WTC showed statistically significant differences when compared to the baseline model (χ^2 (2) = 7.232, p = .027). With respect to the "learners' L2 proficiency" moderator, two paths – one between L2 anxiety and L2 WTC, and another between perceived L2 competence and L2 use – showed statistically significant differences when compared to the baseline model (χ^2 (2) = 14.533, p = .001).

Table 4. Direct, Indirect, and Total Effects of the Constructs in Baseline MASEM Model.

Variables		**Direct effects**			**Indirect effects**			**Total effects**		
IV	**DV**	**Est.**	**SE**	**Z**	**Est.**	**SE**	**Z**	**Est.**	**SE**	**Z**
COM	WTC	.357***	0.040	8.983	–	–	–	.357***	0.040	8.983
	USE	.362*	0.144	2.505	.064	0.070	0.912	.426***	0.113	3.762
ANX	WTC	–.114**	0.036	−3.187	–	–	–	–.114**	0.036	−3.187
	USE	–	–	–	–.020	0.069	−0.296	–.020	0.069	−0.296
MOT	WTC	.187***	0.024	7.501	–	–	–	.187***	0.024	7.501
	USE	.179**	0.070	2.546	.033	0.066	0.505	.212**	0.068	3.124
WTC	USE	.178*	0.090	1.980	–	–	–	.178*	0.090	1.980

*** p < .001, ** p < .01, * p < .05.

Table 5. Results of Moderator Analyses for Five Moderating Variables.

	Overall moderation effects		**Significant moderation effects**				
Moderators	**Y/N**	**Est.**	**Paths**	**Values of Moderators**	**Est.**	**z**	***p*-value**
Learning context	N	χ^2 (6) = 9.863, p = .131	N/A	SL	N/A	N/A	N/A
				FL			
L1-L2 relation	N	χ^2 (6) = 11.663, p = .070	N/A	Related	N/A	N/A	N/A
				Not related			
WTC context	Y	χ^2 (6) = 11.163, p = .025	ANX → WTC	Inside classrooms	–.193	2.060	0.039
		→ χ^2 (2) = 7.232, p = .027		Outside classrooms	–.057		
			COM → WTC	Inside classrooms	.236	2.903	0.003
				Outside classrooms	.429		
Age	N	χ^2 (6) = 10.982, p = .089	N/A	Children	N/A	N/A	N/A
				Adolescents/Adults			
L2 proficiency	Y	χ^2 (6) = 14.963, p = .021	ANX → WTC	Basic	–.275	−2.234	0.025
		→ χ^2 (2) = 14.533, p = .001		Beyond basic	–.113		
			COM → USE	Basic	.644	3.211	0.001
				Beyond basic	.222		

For the third step, we identified to what extent each path coefficient was moderated by the values of the moderator variables. As Figure 5 illustrates, the results indicated that for the "WTC context" moderator, learners' anxiety played a more important role inside classrooms (β = –.193) than outside classrooms (β = –.057), and this difference was statistically significant (z = 2.060, p = .039). In contrast, learners' perceived competence was more important outside classrooms (β = .429) than inside classrooms (β = .236; z = 2.903, p = .003).

With regard to the "learners' L2 proficiency" moderator, as shown in Figure 6, the results indicated that learners' anxiety played a more important role for basic learners (β = –.275) than more proficient learners (β = –.113), and this difference was statistically significant (z = –.2.234, p = .025). Along the line, learners' perceived competence was more important for basic learners (β = .644) than more proficient learners (β = .222; z = 3.211, p = .001).

Discussion

Overall Findings

This study synthesized previous empirical findings from 44 samples (N = 12,094) on the correlations between L2 WTC and its high-evidence factors by constructing a pooled correlation matrix to fit a structural equation model using the MASEM approach. Basically, the results confirmed the findings of Shirvan *et al.* (2019). The pooled correlation coefficients for WTC and motivation (r = .336), anxiety (r = –.311), and perceived competence (r = .467) were nearly identical to those of Shirvan *et al.* (2019; rs = .371, –.292, .480, respectively). What makes the present study distinguishable from Shirvan *et al.* (2019) is that we were able to statistically verify MacIntyre *et al.*'s (1998) heuristic model of L2 WTC, and to examine structural relationships among the variables. In other words, we established evidence-based causal relationships by using the MASEM approach such that L2 motivation and competence positively influenced L2 WTC, which ultimately affects L2 use, while L2 anxiety, which was negatively correlated with competence, negatively affected L2 WTC. These findings were in line with previous SEM studies on this topic (e.g., Hashimoto, 2002; Peng & Woodrow, 2010; Yashima, 2002). More specifically, our structured dataset and the computed estimates (i.e., direct, indirect, and total effects of constructs; Table 5) from the MASEM approach statistically validated the following causal relationships, which have been theoretically supported: (a) L2 competence as well as L2 motivation directly affected L2 WTC (βs = .357 and .187), and directly and indirectly (via L2 WTC) influenced L2 use (βs = .426 and .212); and (b) L2 anxiety directly influenced L2 WTC in a negative way (β = –.114), but did not directly or indirectly affect L2 use (βs = N/A and –.020; not significant).

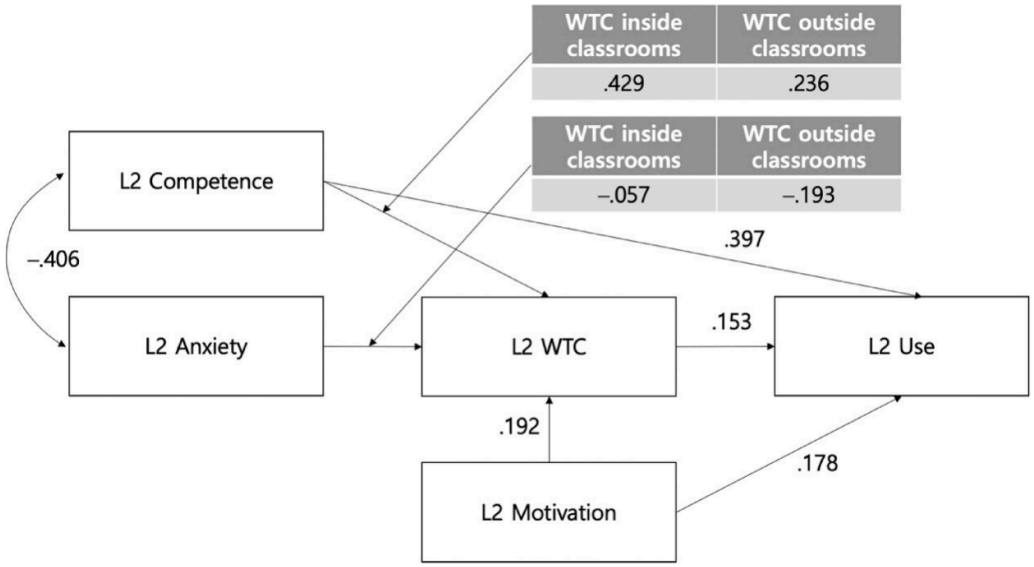


Figure 5. MASEM model with the WTC context (WTC inside classrooms vs. WTC outside classrooms) as a moderator.

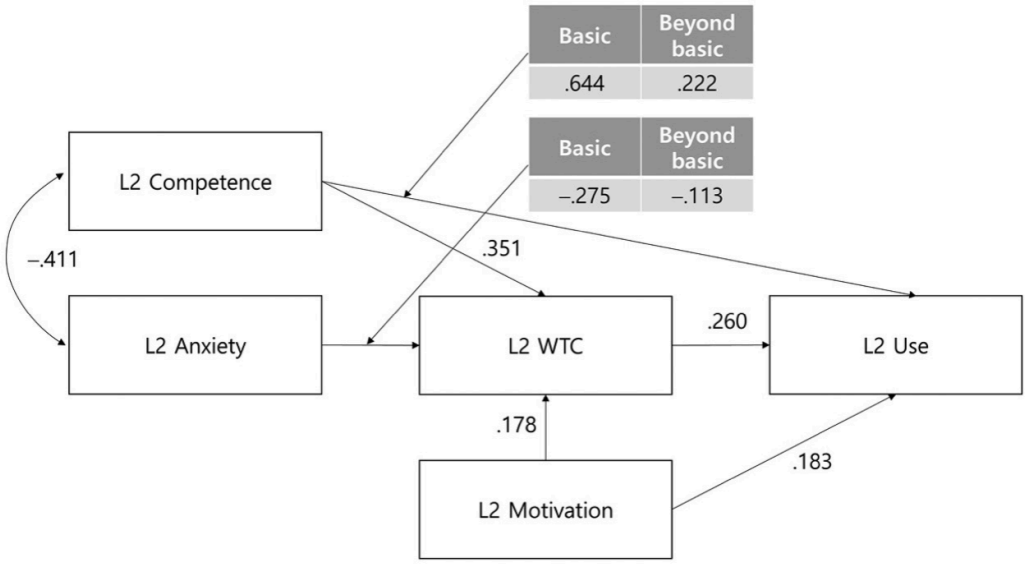


Figure 6. MASEM model with the L2 proficiency (basic vs. beyond basic) as a moderator.

In addition to our successful validation of MacIntyre *et al.*'s (1998) heuristic model of L2 WTC, we were able to statistically confirm that learners' L2 WTC significantly impacts L2 use frequency. Considering the significance of achieving the ultimate goal of L2 learning (i.e., frequent L2 use), particularly from the communicative language teaching (CLT) perspective (Khajavy *et al.*, 2016), this finding demonstrated the role of WTC as a gateway for a variety of individual, situational, and contextual factors to positively or negatively affect L2 learners' decision to willingly enter into L2 communication (MacIntyre *et al.*, 2002; Yashima *et al.*, 2004). This supports scholars and practitioners in the relevant fields, who aim to find theoretical and practical approaches to improve L2 learners' WTC in order to facilitate their use or frequency of communication in L2.

Furthermore, this study quantified the relative importance of the factors in influencing learners' L2 use via WTC, which could not be examined in previous meta-analytic studies. For instance, L2 use was found to be directly influenced only by WTC in some studies (Lin, 2019; Turner *et al.*, 2021; Yashima *et al.*, 2004). In other studies, L2 use was found to be directly influenced by motivation and WTC (Hashimoto, 2002; MacIntyre & Charos, 1996). However, the present study demonstrated that the factors had different degrees of direct and indirect contributions to L2 use from a comprehensive perspective. The MASEM approach revealed the total effects of the factors on learners' L2 use. The results indicated the importance of learners' perceived L2 competence for L2 WTC and *eventually* for L2 use frequency, corresponding to previous findings indicating that L2 competence was the most significant predictor of WTC (Ghonsooly *et al.*, 2012; Hashimoto, 2002, MacIntyre & Charos, 1996; Peng & Woodrow, 2010; Yashima, 2002). As we could identify the statistically significant negative correlation between learners' L2 competence and L2 anxiety, our findings indicate the importance of L2 competence as well as that of L2 anxiety.

In summary, we believe that the most significant finding of the present study was the establishment of the combined role of learners' L2 competence and L2 anxiety in determining one's L2 WTC and L2 use, which we turn to next.

Social Psychological Dimensions of L2 WTC and Pedagogical Implications

One feature of the results of the baseline MASEM model as well as those of the moderator analyses have in common is the finding that learners' L2 competence and L2 anxiety are the core constructs that influence L2 WTC and L2 use. As mentioned above, we identified that learners' L2 competence was the strongest factor in predicting their L2 WTC and L2 use, and that learners' L2 anxiety was not only another significant factor but also closely (negatively) related to L2 competence. We also found that two variables—WTC context and learners' L2 proficiency—had significant moderating effects on the contribution of these two factors to L2 WTC and L2 use. Overall, these findings provide evidence for the importance of social psychological dimensions of L2 WTC as well as pedagogical implications thereof. In doing so, the findings of the present study are expected to (a) illuminate the dynamics in which learners' L2 WTC and L2 use are significantly affected by two social psychological factors, i.e., perceived competence and anxiety; and (b) provide a pedagogical background for diverse educational efforts to increase learners' L2 frequency as well as their L2 WTC by improving L2 competence and lowering L2 anxiety.

Social Psychological Dimensions of L2 WTC. We believe that a social psychological approach would provide a more relevant interpretation as well as insights on the mechanism of L2 WTC and L2 use, particularly for the role of perceived L2 competence and L2 anxiety, which were

suggested to be influential in determining L2 WTC and L2 use. For example, when it comes to WTC contexts (inside vs. outside classrooms), it was found that the impact of learners' L2 anxiety was larger and the role of learners' perceived L2 competence was found to be less important in determining learners' WTC inside classrooms when compared to their WTC outside classrooms. We believe that this may arise from the contextual differences between classroom and out-of-class settings. As Lee and Lee (2020) suggested, it can be inferred that L2 learners are likely to be more nervous or reluctant to participate in L2 communication in a classroom environment than any communicative contexts outside the classroom. In other words, L2 learners may tend to feel more anxious depending on various social and contextual factors inside the classroom: teacher support, student cohesiveness, and task orientation (Peng & Woodrow, 2010), to name a few. Conversely, outside the classroom, L2 learners may feel less nervous since they are able to communicate with people in L2 more naturally, and other factors, such as intergroup relations (Ward & Hewstone, 1985), social distance (Nguyen & Hamid, 2019), and ethnolinguistic vitality (Bourhis *et al.*, 2019; Smith *et al.*, 2018) are more influential than in the classroom.

Likewise, from a social-developmental approach, we could argue that learners are prone to L2 anxiety in the classroom since they may feel pressured by their teachers to demonstrate their ability to communicate in L2 or nervous about speaking in front of their classmates. In addition, the less influential role of perceived competence on L2 WTC as opposed to the influence of L2 anxiety in the classroom can be understood given a variety of potential social psychological, cognitive, and contextual sources of L2 anxiety (Hashimoto, 2002; Khajavy *et al.*, 2016; Horwitz, 2016). In other words, it is plausible that L2 learners who develop perceived competence reduce its influence on L2 WTC while leaving L2 anxiety to play a relatively more prominent role in L2 WTC (Cha & Kim, 2013; Denies *et al.*, 2015; MacIntyre & Charos, 1996).

In addition, when it comes to learners' L2 proficiency, it was identified that students with low L2 proficiency required special attention in that their perceived L2 competence played a larger role in influencing their L2 use, and their L2 anxiety had a greater impact on L2 WTC than in the case of their high proficient peers. From a social psychological perspective, this may be due to L2 learners' inclination to initiate L2 communication that changes more drastically depending on the level of anxiety in using L2 among basic L2 learners than advanced learners (Cetinkaya, 2005; Hashimoto, 2002; MacIntyre *et al.*, 2002). Furthermore, basic L2 learners with higher L2 competence would tend to engage in L2 communication more frequently than those with lower L2 competence (Hashimoto, 2002); this tendency resonates with MacIntyre and Charos' (1996) finding that perceived L2 competence is the strongest predictor of L2 communication frequency of L2 learners who took an introductory L2 conversation class.

These two tendencies of learners with low L2 proficiency may be understood given that they are more vulnerable to various social psychological factors and different types of communication contexts than those with high L2 proficiency; basic L2 learners, therefore, tend to demonstrate more drastic change in their L2 WTC and L2 use according to their perceived L2 competence and L2 anxiety. Overall, we believe that this finding is in line with the general tendency of low-proficiency learners, who have lower competence (MacIntyre *et al.*, 1997) and higher anxiety levels than high-proficiency learners (Gardner & MacIntyre, 1993; Hashimoto, 2002; Yashima, 2002).

Pedagogical Implications of Social Psychological Dimensions of L2 WTC. The discussion of the social psychological dimensions of L2 WTC above will be more meaningful if it can be linked to its pedagogical implications. In other words, the dynamics of perceived L2 competence and L2 anxiety with regard to learners' L2 WTC and L2 use can be beneficial to the stakeholders if it comes with pedagogical discussions for diverse educational efforts to improve L2 competence and lower anxiety to increase learners' L2 use frequency as well as their L2 WTC. For example, from a pedagogical perspective, the important role of L2 anxiety in affecting L2 WTC inside classrooms suggests that teachers should develop and utilize diverse educational methods and pedagogical approaches to help L2 learners lower their anxiety and feel relaxed inside classrooms (Khajavy *et al.*, 2016). So, teachers could make use of a variety of engaging and intriguing L2 activities and tasks (Cao, 2011; Dewaele *et al.*, 2018; Kang, 2005), or create a classroom environment that is supportive and enjoyable for L2 learners (Dörnyei, 2007). Furthermore, the influential role of L2 competence on L2 WTC outside the classroom indicates that L2 learners' WTC can be improved by enhancing their perceived communicative competence, especially by using various pedagogical strategies to make them feel more confident in using L2 in a variety of communicative context beyond the classroom (e.g., Hashimoto, 2002; Khajavy *et al.*, 2016; Pyun *et al.*, 2014), which can ultimately lead to more frequent use of L2.

In addition, the direct influence of perceived L2 competence on L2 use among beginners in a pedagogical environment suggests that teachers should make basic L2 learners feel that they are competent and proficient in L2 use and raise their self-confidence in L2 communication by utilizing various teaching skills and strategies. For example, encouraging basic L2 learners to evaluate their L2 communication ability in a positive way would be a plausible means of raising their perceived competence in L2, which ultimately leads to better L2 outcomes including frequency of L2 use (Gardner, 1985). Similarly, training basic L2 learners to control their own negative feelings toward L2 communication could be another pedagogically sound approach to preparing learners for a variety of communicative opportunities that they will encounter in the course of improving their L2 proficiency. Likewise, we would like to

highlight the importance of providing low-proficiency L2 learners with emotional support, creating a friendly atmosphere in which they do not feel intimidated to engage in L2 communication, and helping them actively change negative mindsets to positive ones (Lou & Noels, 2016, 2019).

To sum up, in order to ultimately help L2 learners to be sufficiently willing to communicate in and actively use their L2 frequently, appropriate pedagogical ways to increase their perceived L2 competence and lower their L2 anxiety should be implemented according to whether L2 learners are situated within or beyond the classroom, and whether they are beginners or advanced L2 learners. In addition, and most importantly, those pedagogical means to enhance learners' willingness and frequency to communicate in L2 should be undergirded by the social psychological dimensions of L2 WTC, in which L2 WTC is at the center of a complex and dynamic nature of various social psychological factors that interact with one another depending on a multitude of communicative contexts.

Conclusion: Summary, Limitations, and Future Directions

In the present study, we built a MASEM model by synthesizing empirical findings from 44 independent samples (N = 12,094) to investigate the structural relationships among WTC and three high-evidence factors in L2 learning contexts. We found that learners' perceived L2 competence and L2 anxiety played a significant role in determining L2 WTC and use. Furthermore, the moderator analyses revealed that the impact of L2 competence and L2 anxiety on L2 WTC was significantly different, depending on L2 WTC contexts (inside vs. outside the classroom) as well as learners' L2 proficiency levels (basic vs. beyond basic). We believe that by providing a comprehensive understanding of the complex social psychological dimensions of L2 WTC, along with evidence-based importance of perceived L2 competence and L2 anxiety, the current study will not only contribute to the literature on L2 WTC but will also serve as a basis for future empirical studies that seek to further develop the proposed L2 WTC framework.

This study was not without any limitation. First, future studies should include more correlation coefficients for the MASEM approach to acquire a more comprehensive L2 WTC model with acceptable model fits. This study included only three high-evidence factors as well as WTC and L2 use. Using more samples would allow the inclusion of other variables, including integrativeness and attitude toward learning situations, such as teachers, learning activities (MacIntyre & Charos, 1996). As path coefficients of variables in a structural equation model are dependent on the other variables in the model (Kline, 2015; Lee *et al.*, 2020), adding more

variables may alter the computed estimates in the present study, and this would provide a more accurate interpretation of the complex relationships among another set of high-evidence factors.

Furthermore, a larger sample would allow the inclusion of more relevant moderators in the MASEM model to further investigate the influence of differences between studies on the overall findings of the baseline model. For instance, students may differ aspects such as working memory capacities (e.g., Lee *et al.*, 2020) and socioeconomic status (e.g., Lee & Lee, 2021), in addition to the currently examined moderators.

Second, in the process of synthesizing raw correlation coefficients from previous literature and building a comprehensive dataset by placing correlation coefficients into the pooled correlation matrix, this study could not completely exclude subjectivity in selecting measurements for target variables and/or computing an average correlation coefficient from multiple measurements. More specifically, we had to combine and average multiple correlation coefficients between constructs of interest, and it is necessary to consider the potential interpretive difficulties of combining and averaging across different measurement types with different emphases and psychometric properties. For example, in terms of motivation, we combined various constructs (e.g., instrumental, integrative, intrinsic, extrinsic motivation) and averaged their correlation coefficients with L2 WTC for practical reasons. In addition, with respect to L2 WTC, while WTC scales were categorized into two types (McCroskey's WTC scales vs. others) for the purpose of the present study, a wider range of various WTC scales have been used across the collected primary studies. This approach was possible because the combined and averaged constructs had similar sizes and directions of correlation coefficients, and we believed that our decision would not seriously distort the collected data. Practically speaking, it is almost impossible to encompass and investigate every different theory and definition of the constructs, and we believe the present study was able to reach a comprehensive understanding of the mechanism between L2 WTC and other related variables affecting L2 WTC in a broad sense. Nevertheless, we strongly hope that future studies include larger samples so that the specification of target variables and inclusion of different types of motivation without losing important pedagogical differences between constructs can be more plausible. Furthermore, we suggest that future studies should include more empirical investigations and conduct follow-up meta-analyses to accumulate findings for a more reliable and accurate research synthesis.

Lastly, we call for future studies to investigate the effects of various pedagogical efforts in order to provide concrete causal evidence for improving L2 WTC. For example, we could conjecture that creating a positive learning environment may contribute to the enhancement of L2 WTC by mitigating learners' anxiety levels (Dewaele, 2019; Dewaele & Dewaele, 2018);

improving learners' growth mindsets (Lou & Noels, 2016, 2019) may also help to lower anxiety in L2 communication and improve WTC, and enhancing L2 learners' autonomy in language learning (Zarrinabadi *et al.*, 2021) may increase their perceived competence, which would directly and indirectly contribute to an increase in L2 WTC and L2 use frequency.

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Supplemental Material

Supplemental material for this article is available online.

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